

Garage: GPU particle-based AR content for futuristic experiences

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Figure 1: (a) Battle with dragon (b) Player (c) In-house HMD

ABSTRACT

In futuristic AR, we believe that both real and virtual objects surrounding the player should be interactive and controllable. With this in mind, we introduce "Garage", which is an AR project based on our particle-based system capable of outputting a variety of interactive content. In our system, everything in the player's surrounding environment is presented as particles and adequately simulated in real-time on a desktop PC equipped with a discrete GPU and rendered to in-house HMD, while their color and depth information is captured and sent by RGB camera and LiDAR sensor. LiDAR data is converted to particles using an algorithm that calculates world coordinated positions from depth values. Our system content and provided experiences have time flexibility and spatial control features that cannot be achieved in conventional polygon mesh-based systems. Additionally, the AR information output is presented in an appealing visual style that conserves GPU resources. Because of these features, Garage has a profound impact on AR immersion, and can augment our perception of the world in a limitless and creative manner. To demonstrate the abilities of our system and build new interactive AR experiences, in this project, we designed a game and created several demos that can give players interesting glimpses into the future world of augmented reality.

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Figure 2: Body splitting and teleportation

1 THE GARAGE

1.1 Spatial control

The first special feature of the Garage system is seen when 3D AR objects encounter objects in the player's surrounding environment. For example, in most modern AR applications, collision information is prepared in advance and then acts to peremptorily interrupt AR object movements. However, a more realistic experience would be

possible if the player's surroundings were also visually affected by such collisions, especially when the objects are massive. Accordingly, we simulate scattering at the collision point and add various other reactions such as size or color changes, as shown in Figure 1(a).

These changes have the potential to intensify object manifestation and facilitate player immersion into the AR world. For example, in our game system, players can control and witness a scene in which a whale bursts from the floor as if it has ascended from the sea surface accompanied by splashing waves; or a magnificent roaring dragon flying around a room while blowing fire and knocking over desks and chairs — all while sitting in the comfort of their homes.

This is accomplished by GPU simulations that provide the appropriate velocity, lifetime, color, angle, and size to millions of separate particles in response to AR object movements. In another example, players can role-play as a mage and fight against dragons while seeing wind gusts raised by their wings. Our game "magic" can also generate sharp needles or lances made from the objects in the player's environment, which further stimulates the sensation that an actual battle is occurring.

We also show various other possibilities, such as demonstrations that enable players to assert control in various ways such as melting objects with laser energy or floating them down a river, while enabling arbitrary control algorithms to produce effects such as audio visualization (Figure 3(a)).



Figure 3: (a) Audio visualization (b) Invisible power

1.2 Time control

Our system is also capable of showing content that mixes past and present space by storing RGB-D information for a certain period and then replaying it at appropriate times. In particular, the ability to reproduce human body parts enables us to produce some futuristic body effects. For example, players can experience unique representations such as experiencing the superpower of teleporting his or her arm (Figure 2) or being invisible (Figure 3(b)).

In another demo created using our method, we constructed a shadowboxing one that enables a player to spar with his or her past self instead of an imaginary opponent (Figure 4) and found that the five-second-ago "self" made some surprisingly unexpected movements in fighting situations. This means that time control features also have the potential to act as sources of unknown movement that can be given to non-player characters.

1.3 Particle styles

Our system is capable of producing high levels of visual flexibility. For example, natural photorealism can be achieved by minimizing



Figure 4: Sparring with past self

particle sizes and sampling captured texture colors to produce "as is" effects. However, noting that world-famous games like Minecraft have shown that game worlds constructed from rough 3D objects can still be appealing, we conjectured that a similar style could be effective in AR because it would impart a sensation of uniformity between virtual and physical objects. Hence, a feature of our system allows the game designer to also choose voxel style.

We also obtain two performance benefits from this style, both of which conserve significant amounts of GPU resources. First, it allows larger particle sizes than when a photorealistic style is used. Second, in cases that involve recording and replaying past frames, the stored RGB-D texture data tend to pile up as recording times lengthen. However, the negative impact is insignificant when voxel style reduced resolution is applied.

Particle sizing is also important because, when designing an AR demo or product, the only way to find the most appropriate particle size is to observe how different sizes affect content impressions. Accordingly, we implemented an interactive tool that enables designers to dynamically change particle size while always maintaining alignment (Figure 5). This enables us to rapidly seek out the most appealing voxel style depending on the situation.

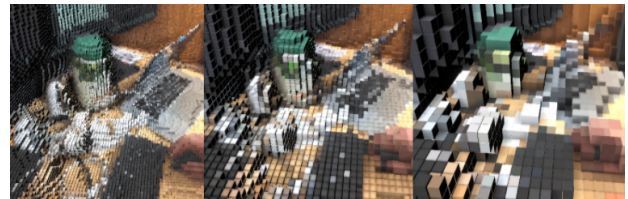


Figure 5: Particle sizing and alignment

2 CONCLUSION AND FUTURE WORK

We have built new interactive AR experiences that cannot be produced by conventional systems because its geometric features and visual effects are boosted by millions of GPU-produced particles. As a result, players can enjoy levels of AR immersion that they have never experienced before.

In the future, we intend to further develop semantic segmentation for discriminating the attributes of surrounding environments and add more accurate simulations optimized for each material. In this manner, the virtual and physical worlds can be expected to become increasingly mixed.